## Iteration 2

Update values : $= \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$ $b = 0.4.318$

① $\hat{y} = Xm + b.1$

$$= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3316 \\ 1.808 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 3.1426 \\ 7.4184 \end{bmatrix}$$

② $e = \hat{y} - y = \begin{bmatrix} 3.1426 \\ 7.4184 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

$$= \begin{bmatrix} -1.8574 \\ 1.4184 \end{bmatrix}$$

③ $Cost \Rightarrow J = \frac{1}{n}(e_1^2 + e_2^2)$

$$= \frac{1}{2}\left((-1.8574)^2 + (1.4184)^2\right)$$

$$= 2.7238$$

(4) Gradient $dJ/dm = X^T e = \begin{bmatrix} 1(1.8574) + 4(1.4134) \\ 10(1.4134) + 3(-1.8574) \end{bmatrix}$

$$\frac{dJ}{dm} = \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

$$\frac{dJ}{db} = 1^T e = -1.8574 + 1.4134$$

$$= -0.444$$

(5) $\alpha = 0.01$

$$m = m - \alpha \frac{dJ}{dm}$$

$$= \begin{bmatrix} -1.3316 \\ 1.1908 \end{bmatrix} - 0.01 \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

$$= \begin{bmatrix} -1.356956^2 \\ 1.095185 \end{bmatrix}$$

$$b = b - \alpha \frac{dJ}{db}$$

$$= 0.9318 - 0.01(0.444)$$

$$0.95624$$

for new cost

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.369562 \\ 1.095185 \end{bmatrix} + \begin{bmatrix} 0.93624 \\ 0.9362 4 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 2.852224 \\ 6.409812 \end{bmatrix}$$

$$e = \hat{y} - y = \begin{bmatrix} 2.852224 \\ 6.409812 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} -2.147776 \\ 0.409812 \end{bmatrix}$$

$$J = \frac{1}{n}(e_1^2 + e_2^2) = \frac{1}{2}\left( (-2.147776)^2 + (0.409812)^2 \right)$$

$$= 2.3904$$